Toward a Non-Commutative Residuated Lattice from Quaternion Multiplication

J. Arturo Ornelas Brand
Independent Researcher, Aguascalientes, México
arturoornelas62@gmail.com

Draft — April 13, 2026

Abstract

The quaternionic G-lattice introduced in the companion paper uses component-wise min as its monoidal operation, making the residuated lattice a product of four independent chains. The quaternionic structure of the framework manifests in the SU(2) group action and the product on S^3, but not in the logic's conjunction. This paper investigates whether the Hamilton product of quaternions can serve as a genuinely non-commutative conjunction—a monoidal operation that couples the four epistemic axes through the quaternionic cross terms.

We establish positive results: the Hamilton product on $V = [0, 1] \times [-1, 1]^3$ satisfies bilateral identity, bilateral annihilation, and associativity; it defines left and right implications via quaternionic division, both satisfying modus ponens; and under every commutative restriction (Boolean, fuzzy, modal), the two implications coincide and the product reduces to the standard commutative t-norm. Non-commutativity is proven to be *emergent*: it arises only in the full four-dimensional space, not in any proper restriction.

We also characterize fundamental obstructions. The Hamilton product is not closed on V and is not monotone with respect to the component-wise order. We prove that no total order on $\mathbb{H}$ is compatible with Hamilton multiplication, that the component-wise lattice order and the Hamilton monoid cannot coexist on V, and—via four independent impossibility results—that no non-trivial lattice order on $B^4 = \{q \in \mathbb{H} : |q| \leq 1\}$ is compatible with the Hamilton monoid. The four results are: (i) no proper bi-invariant cone exists in $\mathbb{H}$ (SU(2) transitivity), (ii) no non-trivial bi-invariant partial order exists on Q_8 (exhaustive closure), (iii) no real-dominant subcone is closed under multiplication (analytical bound), and (iv) the norm order collapses the structure to a one-dimensional product BL-algebra.

These results culminate in a **quaternionic trilemma**: for any monoidal operation on a subset of $V = [0, 1] \times [-1, 1]^3$, at most two of {associativity, non-commutativity, compatible lattice order} can hold simultaneously. The trilemma is a *theorem* for V-domains (Theorem 40): De Moivre's theorem shows that every non-real element's iterates eventually exit V, so the only sub-monoids of V under Hamilton product are commutative. For the full ball B^4, where powers remain bounded but the angular argument fails, Theorem 42 proves that the canonical non-commutative sub-monoid (equal-norm generators) admits no compatible lattice order. The equal-norm hypothesis is sharp: for generators of unequal norm, compatible monotone lattice orders exist (Remark 33).

An α-parameterized family interpolates between the diagonal product ($\alpha = 0$) and the full Hamilton product ($\alpha = 1$), preserving all specializations; associativity holds only at the endpoints.

Keywords: non-commutative residuated lattice; quaternion multiplication; t-norm; substructural logic; fuzzy logic; Hamilton product; non-commutative conjunction

1 Introduction

The quaternionic logic developed in Ornelas Brand [2026a] assigns to each semantic state a four-dimensional coordinate $(r, i, j, k) \in V = [0, 1] \times [-1, 1]^3$, where r encodes crystallised truth, i encodes potentiality, j encodes verifiability, and k encodes recursive selection. Logical connectives form a bounded residuated lattice on V, and an SU(2) group acts by conjugation, yielding a G-lattice whose central result is a commutation theorem: connectives commute with the group action if and only if they depend exclusively on the real component.

The lattice connectives—AND = min, OR = max, and the Gödel residuum—are defined component-wise. The resulting lattice is isomorphic to the product of four independent chains [Ornelas Brand, 2026a, Remark 5]. The quaternionic structure manifests in the group action and the product on S^3, but not in the conjunction itself.

This is a structural limitation. In the four-dimensional space, the Hamilton product couples all four components: the relation $ij = k$ (potentiality composed with verification yields selection) is a cross-term that no component-wise operation can reproduce. If the conjunction were the Hamilton product rather than component-wise min, the logic would be genuinely non-commutative—the order of premises would matter, as in Lambek calculus [Lambek, 1958] and substructural logics [Restall, 2000].

This paper investigates whether such a construction is possible. The question is: *can a residuated lattice be built on V whose monoidal operation derives from quaternion multiplication?*

The answer is: partially. We establish the monoid and its residuals, characterize the obstructions to the full lattice, and map the path forward.

Contributions.

1. The Hamilton product on V satisfies bilateral identity, bilateral annihilation, and associativity (Theorem 1).
2. Two obstructions are characterized: failure of closure (Theorem 4) and failure of monotonicity (Theorem 5), both traced to the signed structure of the quaternionic cross terms.
3. The unit ball B^4 resolves closure (Proposition 7).
4. Left and right implications via quaternionic division satisfy modus ponens, tautology, and non-commutativity (Theorem 12).
5. Under every commutative restriction (Boolean, fuzzy, modal), the two implications coincide and the product reduces to the standard t-norm (Theorem 15).
6. Non-commutativity is emergent: it arises only in the full 4D space (Theorem 18).
7. No total order on $\mathbb{H}$ is compatible with Hamilton multiplication (Proposition 21).
8. Six independent impossibility results rule out every natural candidate for a compatible lattice order on B^4: no bi-invariant cone (Theorem 24), no bi-invariant order on Q_8 (Theorem 25), no closed real-dominant subcone (Proposition 26), norm-order collapse to one dimension (Proposition 27), component-wise incompatibility (Theorem 22), and sphere exclusion for integral RLs (Proposition 28).
9. The **quaternionic trilemma**, proved for V-domains (Theorem 40) via angular accumulation (Theorem 38): at most two of {associativity, non-commutativity, compatible lattice order} hold simultaneously. Proved for the canonical B^4 sub-monoid with equal-norm generators (Theorem 42); the equal-norm condition is sharp (Remark 33).
10. An α-parameterized family bridges the commutative and non-commutative regimes, with associativity only at the endpoints (Proposition 35).

2 Preliminaries

2.1 Quaternion algebra

The quaternion algebra $\mathbb{H}$ is the four-dimensional real algebra spanned by $\{1, i, j, k\}$ with $i^2 = j^2 = k^2 = ijk = -1$. Every quaternion $q = a + bi + cj + dk$ has real part $\mathrm{Re}(q) = a$, imaginary part $\mathrm{Im}(q) = (b, c, d)$, conjugate $\bar{q} = a - bi - cj - dk$, and norm $|q| = \sqrt{a^2 + b^2 + c^2 + d^2}$. The norm is multiplicative: $|q_1 q_2| = |q_1| \cdot |q_2|$.

2.2 Hamilton product

For $q_1 = (r_1, i_1, j_1, k_1)$ and $q_2 = (r_2, i_2, j_2, k_2)$, the Hamilton product is:

$$q_1 \cdot q_2 = \begin{pmatrix} r_1 r_2 - i_1 i_2 - j_1 j_2 - k_1 k_2 \\ r_1 i_2 + i_1 r_2 + j_1 k_2 - k_1 j_2 \\ r_1 j_2 - i_1 k_2 + j_1 r_2 + k_1 i_2 \\ r_1 k_2 + i_1 j_2 - j_1 i_2 + k_1 r_2 \end{pmatrix} \tag{1}$$

The product is associative, has identity $1 = (1, 0, 0, 0)$, and every nonzero quaternion has an inverse $q^{-1} = \bar{q}/|q|^2$.

2.3 Non-commutative residuated lattices

A *residuated lattice* [Galatos et al., 2007] is an algebra $(L, \wedge, \vee, \cdot, \backslash, /, e)$ where $(L, \wedge, \vee)$ is a lattice, $(L, \cdot, e)$ is a monoid, and the *residuation law* holds:

$$a \cdot b \leq c \quad \Longleftrightarrow \quad a \leq c/b \quad \Longleftrightarrow \quad b \leq a \backslash c \tag{2}$$

where $\backslash$ is the left residual and $/$ is the right residual. When $\cdot$ is commutative, the two residuals coincide and we write $a \to b$.

In the commutative case, the monoidal operation $\cdot$ is a t-norm: a binary operation on $[0, 1]$ that is commutative, associative, monotone, and has identity 1. The Gödel t-norm (min), the product t-norm ($a \cdot b$), and the Łukasiewicz t-norm ($\max(0, a+b-1)$) are the three basic t-norms [Hájek, 1998, Esteva & Godo, 2001]. The companion paper Ornelas Brand [2026a] uses min on each component, yielding a product of four Gödel chains.

Non-commutative extensions were introduced by Hájek [2003] (pseudo-BL algebras) and studied extensively by Ciungu [2006] and Galatos et al. [2007]. The key difference from the commutative case is the existence of *two* residuals, corresponding to left and right division.

2.4 Quaternionic fuzzy sets

Dai [2023] introduced quaternion-valued membership functions for fuzzy sets, defining intersection and union via component-wise min and max. However, Dai's construction does not define a t-norm, residuum, or any inference mechanism—it remains at the level of set operations.

3 The Hamilton Product as Conjunction

We examine which properties of a monoidal operation the Hamilton product satisfies on $V = [0, 1] \times [-1, 1]^3$.

Theorem 1 (Monoid-like properties)**.** *The Hamilton product on V satisfies:*

1. ***Bilateral identity:*** *$q \cdot \mathbf{1} = \mathbf{1} \cdot q = q$ for all $q \in V$, where $\mathbf{1} = (1, 0, 0, 0)$.*
2. ***Bilateral annihilation:*** *$q \cdot \mathbf{0} = \mathbf{0} \cdot q = \mathbf{0}$ for all $q \in V$, where $\mathbf{0} = (0, 0, 0, 0)$.*
3. ***Associativity:*** *$(q_1 \cdot q_2) \cdot q_3 = q_1 \cdot (q_2 \cdot q_3)$ for all $q_1, q_2, q_3 \in V$.*

Proof. (1) From (1) with $q_2 = (1, 0, 0, 0)$: the real-times-real term gives r_1, all cross terms vanish. The calculation is symmetric.

(2) Every term in (1) contains a factor from q_2; if $q_2 = \mathbf{0}$, all terms vanish. By symmetry of the bilinear structure.

(3) This is the associativity of the quaternion algebra $\mathbb{H}$, which holds for all quaternions, not just those in V. □

The Hamilton product also satisfies the basis relations that give it semantic content:

Proposition 2 (Basis relations)**.** *The pure basis quaternions satisfy:*

$$i \cdot j = k, \qquad j \cdot i = -k, \tag{3}$$
$$j \cdot k = i, \qquad k \cdot j = -i, \tag{4}$$
$$k \cdot i = j, \qquad i \cdot k = -j, \tag{5}$$
$$i^2 = j^2 = k^2 \qquad = -\mathbf{1}. \tag{6}$$

Remark 3 (Semantic reading)**.** *The non-commutativity $ij = k \neq ji = -k$ has a direct semantic interpretation:* imagining first, then verifying *($ij = k$: selection) produces a different result from* verifying first, then imagining *($ji = -k$: anti-selection). The order of epistemic operations matters. This is the algebraic signature of the distinction between hypothesis generation (imagine, then test) and falsification (test, then reimagine).*

However, two critical properties fail:

Theorem 4 (Closure failure)**.** *The Hamilton product is not closed on $V = [0, 1] \times [-1, 1]^3$. Specifically, for $q_1, q_2 \in V$ with $\mathrm{Im}(q_1) \neq 0$ and $\mathrm{Im}(q_2) \neq 0$:*

$$\mathrm{Re}(q_1 \cdot q_2) = r_1 r_2 - (i_1 i_2 + j_1 j_2 + k_1 k_2) \tag{7}$$

which can be negative (violating $r \in [0, 1]$) when $\langle \mathrm{Im}(q_1), \mathrm{Im}(q_2) \rangle > r_1 r_2$.

Proof. Take $q_1 = (0.1, 0.8, 0, 0)$ and $q_2 = (0.1, 0.8, 0, 0)$. Then $\mathrm{Re}(q_1 \cdot q_2) = 0.01 - 0.64 = -0.63 < 0$. Thus $q_1 \cdot q_2 \notin V$. □

Theorem 5 (Monotonicity failure)**.** *The Hamilton product is not monotone with respect to the component-wise order on V. That is, there exist $a, b, c \in V$ with $b \leq c$ (component-wise) but $a \cdot b \not\leq a \cdot c$.*

Proof. The signed cross terms in (1) are the source. Consider the real component of $a \cdot b$:

$$(a \cdot b)_r = a_r b_r - a_i b_i - a_j b_j - a_k b_k$$

Increasing b_i while holding all else fixed *decreases* the real component of the product (due to the $-a_i b_i$ term). But in the component-wise order, increasing b_i means $b' \geq b$. So $(a \cdot b')_r < (a \cdot b)_r$ while $b' \geq b$: left-monotonicity fails on the real component.

Explicitly: let $a = (0.5, 0.5, 0, 0)$, $b = (0.5, 0.1, 0, 0)$, $c = (0.5, 0.9, 0, 0)$. Then $b \leq c$ component-wise, but $(a \cdot b)_r = 0.25 - 0.05 = 0.20$ while $(a \cdot c)_r = 0.25 - 0.45 = -0.20$. So $a \cdot b > a \cdot c$ on the real component. □

Remark 6 (The fundamental obstruction)**.** *Both failures—closure and monotonicity—trace to the same algebraic source: the* ***signed cross terms*** *in the Hamilton product. The real component formula $r_1 r_2 - \langle \mathrm{Im}(q_1), \mathrm{Im}(q_2) \rangle$ contains a subtraction of the imaginary dot product. In a standard (commutative) t-norm on $[0, 1]$, all terms are positive, ensuring closure and monotonicity. In the Hamilton product, the cross terms introduce* negative contributions *that can dominate the positive diagonal terms. This is not a defect—it is the algebraic mechanism that produces non-commutativity and axis coupling ($ij = k$)—but it is incompatible with a positively-ordered bounded domain.*

4 The Unit Ball Monoid

The closure failure on V can be resolved by working on the unit ball.

Proposition 7 (Unit ball monoid)**.** *The set $B^4 = \{q \in \mathbb{H} : |q| \leq 1\}$ with the Hamilton product forms a monoid:*

1. ***Closure**: If $|q_1| \leq 1$ and $|q_2| \leq 1$, then $|q_1 \cdot q_2| = |q_1| \cdot |q_2| \leq 1$.*
2. ***Associativity**: Inherited from $\mathbb{H}$.*
3. ***Identity**: $\mathbf{1} = (1, 0, 0, 0)$ with $|\mathbf{1}| = 1 \in B^4$.*

Proof. Closure follows from the multiplicativity of the quaternion norm. □

Remark 8 (Relation to V)**.** *The domain $V = [0, 1] \times [-1, 1]^3$ and the unit ball B^4 are different subsets of $\mathbb{H}$. Their intersection $V \cap B^4 = \{q \in V : |q| \leq 1\}$ is nonempty and contains $\mathbf{0}$ and $\mathbf{1}$. The unit ball B^4 is closed under the Hamilton product; V is not. However, B^4 is not closed under the lattice operations* min *and* max *(component-wise max of two elements in B^4 can leave B^4), so the lattice and monoid structures live on incompatible domains. This domain incompatibility is the geometric face of the algebraic obstruction identified in Remark 6.*

The unit ball monoid admits left and right division:

Proposition 9 (Division on B^4)**.** *For $a \in B^4$ with $a \neq \mathbf{0}$:*

$$a^{-1} = \bar{a} \,/\, |a|^2 \tag{8}$$

$$a \backslash b = a^{-1} \cdot b \quad \textit{(left division)} \tag{9}$$

$$b / a = b \cdot a^{-1} \quad \textit{(right division)} \tag{10}$$

Left division satisfies $a \cdot (a \backslash b) = b$ and right division satisfies $(b/a) \cdot a = b$.

Proof. By associativity: $a \cdot (a^{-1} \cdot b) = (a \cdot a^{-1}) \cdot b = b$. Similarly for the right division. □

Remark 10 (Division leaves B^4)**.** *For $a, b \in B^4$ with $|a| < |b|$, the division $a \backslash b$ has norm $|b|/|a| > 1$, so $a \backslash b \notin B^4$. This means the residuals of the Hamilton product, defined as suprema via the residuation law (2), do not stay within B^4 in general. A truncation* $\min(1, |b|/|a|)$ *applied to the norm is the natural remedy—analogous to the* $\min(1, b/a)$ *in the product t-norm's residuum on $[0, 1]$—but it breaks the exact adjunction property. Remark 44 characterises the scope of this limitation.*

5 Two Implications from Non-Commutativity

The non-commutativity of the Hamilton product gives rise to two distinct implication operators, corresponding to left and right division. We study their properties on $\mathbb{H} \backslash \{0\}$ (the natural domain where division is defined).

Definition 11 (Left and right implications)**.** *For $a, b \in \mathbb{H}$ with $a \neq \mathbf{0}$:*

$$a \to_L b = a^{-1} \cdot b \quad \textit{(left implication)} \tag{11}$$

$$a \to_R b = b \cdot a^{-1} \quad \textit{(right implication)} \tag{12}$$

Theorem 12 (Properties of quaternionic implications)**.** *Both implications satisfy:*

1. ***Tautology**: $a \to_L a = a \to_R a = \mathbf{1}$ for all $a \neq \mathbf{0}$.*
2. ***Modus ponens (left)**: $a \cdot (a \to_L b) = b$.*
3. ***Modus ponens (right)**: $(a \to_R b) \cdot a = b$.*
4. ***Non-commutativity**: $a \to_L b \neq a \to_R b$ in general.*

Proof. (1) $a^{-1} \cdot a = \mathbf{1}$ and $a \cdot a^{-1} = \mathbf{1}$.

(2) $a \cdot (a^{-1} \cdot b) = (aa^{-1}) \cdot b = b$ by associativity.

(3) $(b \cdot a^{-1}) \cdot a = b \cdot (a^{-1}a) = b$ by associativity.

(4) $a^{-1}b \neq ba^{-1}$ whenever a^{-1} and b do not commute. For example, $i \to_L j = i^{-1} \cdot j = (-i) \cdot j = -k$, while $i \to_R j = j \cdot i^{-1} = j \cdot (-i) = k$. The left and right implications differ by sign on the k axis. □

Remark 13 (Semantic reading of the two implications)**.** *The left implication $a \to_L b = a^{-1}b$ asks: "given a as premise, what must I compose on the right to reach b?" The right implication $a \to_R b = ba^{-1}$ asks: "what must I compose on the left to reach b, given a as premise?" In epistemic terms: the left implication assumes the premise acts first (and asks what follows); the right implication assumes the conclusion is the target (and asks what precedes). The two coincide when the order of operations does not matter—i.e., when the product is commutative.*

Proposition 14 (Contraposition)**.** *Classical contraposition ($a \to b \iff \neg b \to \neg a$) does* not *hold for the quaternionic implications. With conjugation as negation ($\neg q = \bar{q}$):*

$$a \to_L b = a^{-1}b \quad \neq \quad \bar{b} \to_L \bar{a} = \bar{b}^{-1}\bar{a} \tag{13}$$

in general.

Proof. $a^{-1}b = \bar{a}b/|a|^2$ while $\bar{b}^{-1}\bar{a} = b\bar{a}/|b|^2$. These are equal only when $|a| = |b|$ and $\bar{a}b = b\bar{a}$, which fails generically. □

6 Specialization Recovery

A construction on the full four-dimensional space must reduce to known logics under the appropriate restrictions. We verify this for the Hamilton product.

Theorem 15 (Specialization and commutativity)**.** *Under each of the following restrictions, the Hamilton product becomes commutative and reduces to a standard t-norm, and the two implications coincide:*

1. ***Boolean*** *($i = j = k = 0$, $r \in \{0, 1\}$): The product is $r_1 \cdot r_2$ (multiplication in $\{0, 1\}$), which is Boolean AND.*
2. ***Fuzzy*** *($i = j = k = 0$, $r \in [0, 1]$): The product is $r_1 \cdot r_2$ (the product t-norm T_{prod}).*
3. ***Modal*** *($j = k = 0$, $r, i \in [0, 1]$): The product is $(r_1r_2 - i_1i_2,\ r_1i_2 + i_1r_2,\ 0,\ 0)$, which is complex multiplication restricted to the first quadrant. This is commutative.*

In each case, the Hamilton cross terms involving j and k vanish, eliminating the source of non-commutativity.

Proof. (1) With $i_1 = j_1 = k_1 = i_2 = j_2 = k_2 = 0$, equation (1) gives $(r_1r_2, 0, 0, 0)$.

(2) Same calculation with $r_1, r_2 \in [0, 1]$.

(3) With $j_1 = k_1 = j_2 = k_2 = 0$, the cross terms $j_1k_2 - k_1j_2 = 0$, $k_1i_2 - i_1k_2 = 0$, and $i_1j_2 - j_1i_2 = 0$ all vanish. The remaining product is $(r_1r_2 - i_1i_2,\ r_1i_2 + i_1r_2,\ 0,\ 0)$, which is the multiplication of complex numbers $z_1 = r_1 + i_1 \cdot i$ and $z_2 = r_2 + i_2 \cdot i$. Complex multiplication is commutative. □

Corollary 16 (Implications coincide under restriction)**.** *Under any of the three restrictions above, $a \to_L b = a \to_R b$ for all a, b in the restricted domain.*

Proof. Commutativity implies $a^{-1}b = ba^{-1}$. □

Remark 17 (Where non-commutativity begins)**.** *The minimal restriction that produces non-commutativity requires at least two distinct imaginary axes to be active. Specifically, the cross terms in* (1) *are:*

$$j_1k_2 - k_1j_2 \quad \textit{(i-component)}, \quad k_1i_2 - i_1k_2 \quad \textit{(j-component)}, \quad i_1j_2 - j_1i_2 \quad \textit{(k-component)} \tag{14}$$

Each cross term couples two different *imaginary axes. When only one imaginary axis is active (i only, or j only, or k only), all cross terms vanish and the product is commutative. Non-commutativity requires interaction between at least two of* $\{i, j, k\}$.

7 Emergent Non-Commutativity

Theorem 18 (Non-commutativity is emergent)**.** *Let* $S \subsetneq \{r, i, j, k\}$ *be a proper subset of the four axes and* $V_S = \{q \in V : q_m = 0 \text{ for } m \notin S\}$ *the corresponding restriction. If* $|S \cap \{i, j, k\}| \leq 1$*, then the Hamilton product is commutative on* V_S.

Proof. If the active imaginary axes are $\{i\}$, $\{j\}$, or $\{k\}$ (or empty), then all cross terms in (1) vanish (each cross term couples two distinct imaginary axes). The product reduces to real-times-real plus a single imaginary-squared term, which commutes.

For zero imaginary axes ($S \subseteq \{r\}$): the product is r_1r_2, trivially commutative.

For one imaginary axis, say $S = \{r, i\}$: the product is $(r_1r_2 - i_1i_2,\ r_1i_2 + i_1r_2) = z_1z_2$ (complex multiplication), which is commutative. □

Corollary 19. *The Hamilton product is non-commutative on* V_S *if and only if* $|S \cap \{i, j, k\}| \geq 2$.

Proof. The "only if" direction is Theorem 18. For the "if" direction: with two imaginary axes active, at least one cross term is nonzero. For example, with i and j active: $q_1 = (0, a, 0, 0)$, $q_2 = (0, 0, b, 0)$ gives $q_1q_2 = (0, 0, 0, ab)$ while $q_2q_1 = (0, 0, 0, -ab)$. □

Remark 20 (Emergence as dimensional phenomenon)**.** *Non-commutativity is not present in any one- or two-dimensional restriction of the quaternionic logic. It emerges only when at least three dimensions are active (real plus two imaginary). This is analogous to the classical result that the only associative division algebras over* $\mathbb{R}$ *are* $\mathbb{R}$ *(1D, commutative),* $\mathbb{C}$ *(2D, commutative), and* $\mathbb{H}$ *(4D, non-commutative): the non-commutativity is a genuinely four-dimensional phenomenon that cannot be observed from any lower-dimensional shadow.*

In the semantic interpretation: imagining (i), verifying (j), and selecting (k) are each individually order-independent. It is only when two or more *of these processes interact that the order of operations begins to matter.*

8 The Lattice Order Problem

The results so far establish a monoid $(B^4, \cdot, \mathbf{1})$ with left and right residuals on the punctured ball. What is missing for a residuated lattice is a compatible lattice order.

Proposition 21 (No compatible total order)**.** *There is no total order* $\leq$ *on* $\mathbb{H}$ *such that* $(\mathbb{H}, \leq)$ *is a totally ordered group under Hamilton multiplication.*

Proof. In a totally ordered group, every square is non-negative: $a^2 \geq e$ (since either $a \geq e$ or $a^{-1} \geq e$, and $a^2 = a \cdot a \geq a \cdot e = a \geq e$ or $a^2 = (a^{-1})^{-2} \geq e$). But $i^2 = -1 < 1 = e$ in any order that makes e the identity. Contradiction. □

Theorem 22 (Component-wise order incompatibility)**.** *Let* $\leq_V$ *be the component-wise partial order on* V*. There is no subset* $D \subseteq V$ *containing* $\mathbf{0}$ *and* $\mathbf{1}$ *such that* $(D, \leq_V, \cdot_{\mathbb{H}})$ *is a residuated lattice—i.e., such that* D *is closed under* min, max, *and* $\cdot_{\mathbb{H}}$ *simultaneously.*

Proof. Suppose such D exists. Since $\mathbf{0}, \mathbf{1} \in D$ and D is closed under min and max, D contains all elements of the form $(r, 0, 0, 0)$ for $r \in [0, 1]$ (as $\max(\mathbf{0}, \min(\mathbf{1}, (r, 0, 0, 0))) = (r, 0, 0, 0)$, taking D to contain a generating set).

For D to be a non-trivial quaternionic lattice (not merely the real line), D must contain some element $a = (r_a, i_a, 0, 0)$ with $i_a > 0$. Closure under Hamilton product requires $a^2 = (r_a^2 - i_a^2,\ 2r_a i_a,\ 0,\ 0) \in D$. If $i_a > r_a$, then $\mathrm{Re}(a^2) = r_a^2 - i_a^2 < 0$, so $a^2 \notin V$ (the real component is negative). Thus D cannot contain any element with $i_a > r_a$.

But even restricting to $i_a \leq r_a$: closure under max with another element $b = (r_b, i_b, 0, 0)$ gives $\max(a, b) = (\max(r_a, r_b), \max(i_a, i_b), 0, 0)$, which can have $i > r$ (take $r_a < r_b$ and $i_a > i_b$ with $i_a > r_b$). So D cannot be closed under both max and $\cdot_{\mathbb{H}}$ on V. □

Remark 23 (The core tension)**.** *The lattice and the monoid pull in opposite directions:*

- *The* lattice *wants a positively-ordered bounded domain where* min *and* max *are closed.*
- *The* Hamilton monoid *wants a norm-bounded domain where the signed cross terms can roam freely.*

The component-wise order bounds each component independently; the Hamilton product couples them through signed cross terms. The incompatibility is structural: it arises from the minus signs *in the quaternion multiplication table ($i^2 = j^2 = k^2 = -1$), which are the same signs that produce non-commutativity. Removing them would make the product commutative (and component-wise), recovering the existing G-lattice.*

We now strengthen these negative results by ruling out four natural candidate orders on B^4.

Theorem 24 (Cone impossibility)**.** *There is no proper cone $C \subsetneq \mathbb{H} \setminus \{0\}$ that is invariant under both left and right multiplication by unit quaternions. Consequently, no bi-invariant partial order on $\mathbb{H}$ can be defined via a positive cone.*

Proof. The group SU(2) acts on $\mathbb{H}$ by conjugation: $q \mapsto uq\bar{u}$ for $|u| = 1$. This action preserves $\mathrm{Re}(q)$ and acts as SO(3) on $\mathrm{Im}(q)$. Since SO(3) acts transitively on S^2, the conjugation orbit of any pure imaginary unit quaternion is all of $S^2 \subset \mathrm{Im}(\mathbb{H})$.

Suppose C is a cone invariant under conjugation (a necessary condition for bi-invariance). If C contains any q with $\mathrm{Im}(q) \neq 0$, then C contains the entire conjugation orbit of q, which includes elements with imaginary part pointing in every direction of S^2. In particular, C contains both q' and elements with opposite imaginary part, so C contains elements in every open half-space of $\mathbb{H}$. Thus C cannot be a proper cone (a proper cone is contained in a closed half-space).

For left-invariance specifically: if $q \in C$ and $|u| = 1$, then $uq \in C$. But $\{uq : u \in S^3\}$ sweeps out the sphere of radius $|q|$ in $\mathbb{H}$, so C contains the full sphere $S^3_{|q|}$, which is not contained in any proper cone. □

Theorem 25 (Q_8 order impossibility)**.** *No non-trivial bi-invariant partial order exists on the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.*

Proof. Suppose $\leq$ is a partial order on Q_8 such that $a \leq b$ implies $ca \leq cb$ and $ac \leq bc$ for all $c \in Q_8$. Take any $a \neq b$ with $a < b$. Left-multiplying by c gives $ca < cb$; right-multiplying gives $ac < bc$. We generate the *order closure*: the smallest partial order containing $a < b$ and closed under left and right multiplication.

We claim this closure always produces a cycle.

Explicit example: seed $1 < i$. Left-multiply by i: $i < i^2 = -1$. Left-multiply $i < -1$ by i: $-1 < i \cdot (-1) = -i$. Left-multiply $-1 < -i$ by i: $-i < i \cdot (-i) = 1$. The closure now contains $1 < i < -1 < -i < 1$, a cycle contradicting antisymmetry.

General argument: for any $a \neq b$ with $a < b$, left-multiply by $c = ba^{-1}$ (a unit): $ca < cb$, i.e., $b < cb$. The map $x \mapsto cx$ is a permutation of Q_8 of order dividing 4 (every non-identity element of Q_8 has order 4). So iterating gives $b < cb < c^2 b < c^3 b < c^4 b = b$, a contradictory cycle. Exhaustive computation confirms that all 56 ordered pairs (a, b) with $a \neq b$ in Q_8 produce cycles within 4 steps. □

Proposition 26 (Real-dominant subcone non-closure)**.** *For $\tau \in (0,1)$, define the real-dominant subcone $V_\tau = \{q \in B^4 : \mathrm{Re}(q) \geq \tau \cdot |q|\}$. Then V_τ is not closed under the Hamilton product.*

Proof. Take $q_1, q_2 \in S^3 \cap V_\tau$ with $\mathrm{Re}(q_i) = \tau$ and imaginary parts aligned: $\mathrm{Im}(q_1) = \mathrm{Im}(q_2) = v$ with $|v| = \sqrt{1-\tau^2}$. Then:

$$\mathrm{Re}(q_1 q_2) = \tau^2 - |\,\mathrm{Im}(q_1)||\,\mathrm{Im}(q_2)|\cos 0 = \tau^2 - (1-\tau^2) = 2\tau^2 - 1$$

For $q_1 q_2 \in V_\tau$ we need $\mathrm{Re}(q_1 q_2) \geq \tau |q_1 q_2| = \tau$, so we need $2\tau^2 - 1 \geq \tau$, equivalently $2\tau^2 - \tau - 1 \geq 0$, which factors as $(2\tau+1)(\tau-1) \geq 0$. Since $\tau > 0$, this holds only for $\tau \geq 1$. Thus V_τ is not closed for any $\tau < 1$, and V_1 is the real line. □

Proposition 27 (Norm-order collapse)**.** *Define the* norm order *on B^4 by $a \preceq b$ iff $|a| \leq |b|$. Then $(B^4, \preceq, \cdot)$ satisfies:*

1. *Monotonicity: $a \preceq b$ implies $c \cdot a \preceq c \cdot b$ and $a \cdot c \preceq b \cdot c$.*
2. *The quotient $B^4/\!\sim$ (where $a \sim b$ iff $|a| = |b|$) is isomorphic to $([0,1], \leq, \cdot_{\mathrm{prod}})$, the product BL-algebra on $[0,1]$.*

Consequently, the norm order erases all quaternionic structure: the non-commutativity, axis coupling, and four-dimensionality are lost in the quotient.

Proof. (1) follows from $|c \cdot a| = |c||a| \leq |c||b| = |c \cdot b|$.

(2) The map $\pi\colon B^4 \to [0,1]$, $q \mapsto |q|$ is a surjective monoid homomorphism: $\pi(q_1 q_2) = |q_1 q_2| = |q_1||q_2| = \pi(q_1) \cdot \pi(q_2)$, and $\pi(\mathbf{1}) = 1$. It is order-preserving by definition of $\preceq$. The fibers $\pi^{-1}(r)$ are the 3-spheres $r \cdot S^3$ (for $r > 0$), on which $\preceq$ cannot distinguish elements. Thus $B^4/\!\sim \cong [0,1]$ as an ordered monoid, and the product $\cdot_{\mathrm{prod}}$ on $[0,1]$ has the standard residuum $a \to b = \min(1, b/a)$. □

8.1 Sphere exclusion

The integrality condition (monoid identity = lattice top) imposes a further constraint.

Proposition 28 (Sphere exclusion)**.** *Let $(D, \leq, \cdot)$ be an integral residuated lattice with $D \subseteq B^4$, monoid identity $\mathbf{1} = (1,0,0,0)$, and $\mathbf{1}$ as lattice top. If $a \in D \cap S^3$ and $a^{-1} \in D$, then $a = \mathbf{1}$.*

Proof. Since $\mathbf{1}$ is the lattice top, $a \leq \mathbf{1}$ for all $a \in D$. Left-multiply by a^{-1}: $a^{-1} \cdot a \leq a^{-1} \cdot \mathbf{1}$, i.e., $\mathbf{1} \leq a^{-1}$. But $a^{-1} \leq \mathbf{1}$ (top). By antisymmetry, $a^{-1} = \mathbf{1}$, hence $a = \mathbf{1}$. □

Corollary 29. *In any integral residuated lattice on $D \subseteq B^4$ with Hamilton product, if D is closed under inversion on S^3, then $D \cap S^3 = \{\mathbf{1}\}$. The domain can contain at most the identity on the unit sphere; all other elements must have $|q| < 1$.*

8.2 The quaternionic trilemma

The six results above—cone impossibility (Theorem 24), Q_8 impossibility (Theorem 25), subcone non-closure (Proposition 26), norm-order collapse (Proposition 27), component-wise incompatibility (Theorem 22), and sphere exclusion (Proposition 28)—eliminate every natural family of lattice orders on B^4.

Theorem 30 (Quaternionic trilemma)**.** *Let $T = \cdot_{\mathbb{H}}$ be the Hamilton product. Consider the three properties:* (a) *associativity,* (b) *non-commutativity arising from quaternionic cross-term coupling,* (c) *compatible lattice order.*

(I) (V-domains.) *For any $D \subseteq V$ containing $\mathbf{0}$ and $\mathbf{1}$ and closed under T, at most two of* (a)–(c) *hold simultaneously* (*Theorem 40*)*.*

(II) (B^4, equal norms.) *For orthogonal unit purely imaginary $\hat{u}, \hat{v}$ and $r \in (0,1)$, the submonoid $\langle r\hat{u}, r\hat{v}\rangle \cup \{\mathbf{0}, \mathbf{1}\}$ satisfies* (a) *and* (b) *but admits no compatible lattice order* (*Theorem 42*)*.*

The equal-norm condition in (II) *is sharp: when* $|a| \neq |b|$*, compatible monotone lattice orders exist* (*Remark 33*).

Proof. (I) is Theorem 40; (II) is Theorem 42. Sharpness is established by computational verification in Remark 33. □

The six elimination results complement the direct proofs:

Theorem 31 (Elimination of natural orders)**.** *For the Hamilton monoid* $(B^4, \cdot, \mathbf{1})$*, properties (a) and (b) hold, and the following families of orders fail to provide (c):*

1. *Positive-cone orders: no proper bi-invariant cone exists (Theorem 24).*
2. *Finite subgroup orders:* Q_8 *admits no bi-invariant partial order (Theorem 25).*
3. *Subcone orders: no real-dominant subcone* V_τ *with* $\tau < 1$ *is closed (Proposition 26).*
4. *Norm-based orders: the norm order collapses to* $[0, 1]$*, erasing non-commutativity (Proposition 27).*
5. *Component-wise order: incompatible with Hamilton closure (Theorem 22).*
6. *Inversion-closed orders: the domain can meet* S^3 *only at* $\{\mathbf{1}\}$ *(Proposition 28).*

Proof. Each item is proved in the cited result. The theorem collects them to show that six independent approaches to constructing (c) all fail for the Hamilton monoid. □

Remark 32 (Achievable pairs)**.** *Properties (a)+(b) are achieved by the Hamilton product on* B^4*. Properties (a)+(c) are achieved by the G-lattice of Ornelas Brand [2026a] (component-wise min with the standard order on* V*). Properties (b)+(c) are achieved by the Hamilton product on sub-monoids* $\langle r_1\hat{u}, r_2\hat{v}\rangle \cup \{\mathbf{0}, \mathbf{1}\}$ *with* $r_1 \neq r_2$*: the multiplication is non-commutative and a compatible monotone lattice order exists (Remark 33). Thus every pair of properties is individually achievable; only the triple is impossible.*

Remark 33 (Sharpness of the equal-norm condition)**.** *The equal-norm hypothesis* $|a| = |b|$ *in Theorem 42 is essential, not merely a simplifying assumption. When* $|a| = |b| = r$*, the identity* $a^2 = b^2 = -r^2$ *collapses four candidate level-2 products to three* (a^2, ab, ba)*, and the substitution* $a \cdot b^2 = a \cdot a^2 = a^3$ *creates a cyclic swap among level-3 elements that forces a contradiction.*

When $|a| \neq |b|$*,* $a^2 \neq b^2$ *and level 2 has four distinct elements. The chain* $a^3 \leq a^2b \leq ab^2$ *no longer closes to a cycle, because* $ab^2 \neq a^3$*. Computational verification confirms that the sub-monoid* $\langle r_1\hat{u}, r_2\hat{v}\rangle \cup \{\mathbf{0}, \mathbf{1}\}$ *with* $r_1 \neq r_2$ *admits a compatible monotone lattice order at every tested depth (up to depth 5, with 18 elements). The solver finds satisfying assignments in very few backtracking nodes, indicating that the lattice order is not marginally feasible but genuinely unconstrained.*

9 The Alpha Bridge

The tension between the commutative lattice (closed, monotone, no coupling) and the Hamilton product (open, non-monotone, full coupling) suggests a parameterized interpolation.

Definition 34 (Alpha-parameterized product)**.** *For* $\alpha \in [0, 1]$ *and* $q_1, q_2 \in V$*, define:*

$$(q_1 \cdot_\alpha q_2) = \begin{pmatrix} r_1r_2 - \alpha(i_1i_2 + j_1j_2 + k_1k_2) \\ r_1i_2 + i_1r_2 + \alpha(j_1k_2 - k_1j_2) \\ r_1j_2 + j_1r_2 + \alpha(k_1i_2 - i_1k_2) \\ r_1k_2 + k_1r_2 + \alpha(i_1j_2 - j_1i_2) \end{pmatrix} \tag{15}$$

At $\alpha = 0$*, the cross terms vanish and the product becomes:* $(q_1 \cdot_0 q_2)_r = r_1r_2$ *and* $(q_1 \cdot_0 q_2)_m = r_1q_{2,m} + q_{1,m}r_2$ *for* $m \in \{i, j, k\}$*. This is the* diagonal part *of the Hamilton product—the real component multiplies while the imaginary components combine linearly, weighted by the real*

parts. It is not *the component-wise (Hadamard) product $q_{1,m}q_{2,m}$, nor the component-wise min used in the G-lattice of Ornelas Brand [2026a]. At $\alpha = 1$, the product is the full Hamilton product* (1).

Proposition 35 (Properties of the α-family)**.**

1. ***Identity**: $q \cdot_\alpha \mathbf{1} = \mathbf{1} \cdot_\alpha q = q$ for all $\alpha \in [0, 1]$.*
2. ***Annihilator**: $q \cdot_\alpha \mathbf{0} = \mathbf{0} \cdot_\alpha q = \mathbf{0}$ for all α.*
3. ***Associativity**: $\cdot_\alpha$ is associative if and only if $\alpha \in \{0, 1\}$.*
4. ***Commutativity**: $\cdot_\alpha$ is commutative if and only if $\alpha = 0$.*
5. ***Specialization**: Under any restriction that sets two or more imaginary axes to zero, $\cdot_\alpha$ reduces to the same operation for all α (the cross terms vanish).*

Proof. (1)–(2) follow from direct substitution: the cross terms vanish when one factor is $\mathbf{1}$ or $\mathbf{0}$.

(3) At $\alpha = 0$: direct computation gives $((q_1 \cdot_0 q_2) \cdot_0 q_3)_r = (r_1 r_2) r_3 = r_1 (r_2 r_3) = (q_1 \cdot_0 (q_2 \cdot_0 q_3))_r$ and, for each imaginary axis m, $((q_1 \cdot_0 q_2) \cdot_0 q_3)_m = (r_1 r_2) q_{3,m} + (r_1 q_{2,m} + q_{1,m} r_2) r_3 = r_1 r_2 q_{3,m} + r_1 q_{2,m} r_3 + q_{1,m} r_2 r_3 = r_1 (r_2 q_{3,m} + q_{2,m} r_3) + q_{1,m} (r_2 r_3) = (q_1 \cdot_0 (q_2 \cdot_0 q_3))_m$. At $\alpha = 1$: associativity is inherited from $\mathbb{H}$.

For $0 < \alpha < 1$: the α-product is a linear interpolation between two different associative operations. Linear interpolation of associative operations is generically non-associative. To verify: take $a = (0, 0.5, 0, 0)$, $b = (0, 0, 0.5, 0)$, $c = (0.5, 0, 0, 0.5)$ and compute $(a \cdot_\alpha b) \cdot_\alpha c$ vs. $a \cdot_\alpha (b \cdot_\alpha c)$ for $\alpha = 0.5$. The intermediate products generate terms in α^2 that do not cancel.

(4) The cross terms $j_1 k_2 - k_1 j_2$, etc., are antisymmetric in (q_1, q_2). For $\alpha > 0$, they contribute non-symmetrically; for $\alpha = 0$, they vanish.

(5) With two or more imaginary axes zeroed out, every cross term in (15) contains a factor from a zeroed axis and vanishes. □

Remark 36 (Interpretation)**.** *The parameter α controls the degree of **epistemic coupling** between the four axes:*

- *$\alpha = 0$: knowing (r), imagining (i), verifying (j), and selecting (k) are independent processes. The conjunction is component-wise.*
- *$\alpha = 1$: the four processes interact through the full quaternionic structure. Imagining and verifying generate selection ($ij = k$). The order of operations matters.*

The bridge is continuous but not algebraically smooth: associativity holds only at the two endpoints. The loss of associativity for intermediate α means that a gradual transition from commutative to non-commutative logic is not algebraically coherent—the jump must be discrete.

10 Discussion

10.1 Summary of the landscape

Table 1 summarizes which properties are satisfied by each construction.

Table 1: Properties of candidate constructions. The trilemma manifests as the impossibility of all seven checkmarks in any single column.

Property	**Min**/V	**Ham.**/V	**Ham.**/B^4	**Norm-ord.**	α=0	$\alpha\in(0,1)$
Identity	✓	✓	✓	✓	✓	✓
Annihilator	✓	✓	✓	✓	✓	✓
Associative	✓	✓	✓	✓	✓	×
Closed	✓	×	✓	✓	✓	partial
Monotone	✓	×	×	✓	✓	×
Commutative	✓	×	×	eff. ✓	✓	×
Axis coupling	×	✓	✓	×	×	partial

The column "Norm-ord." refers to the norm-order RL of Proposition 27: the Hamilton product is formally non-commutative, but the order quotients it to the commutative $[0, 1]$ product algebra ("effectively commutative").

The table confirms the trilemma: every column with axis coupling (✓ in the last row) lacks monotonicity, and every column with both associativity and monotonicity lacks axis coupling.

10.2 Comparison with known structures

Table 2: The Hamilton product in the taxonomy of monoidal operations.

Structure	**Properties**	**Relation to this work**
t-norm [Hájek, 1998]	Commutative, associative, monotone on $[0, 1]$	Min t-norm is the current G-lattice conjunction
psBL-algebra [Hájek, 2003]	Non-commutative, two residuals on $[0, 1]$	Target structure; our monoid satisfies the monoid axioms but not the lattice axioms
Quantale [Mulvey, 1986]	Complete lattice + associative monoid	The unit ball monoid is a candidate if a compatible lattice order is found
Lambek calculus [Lambek, 1958]	Non-commutative conjunction in linguistics	Shares the two-residual structure; our cross terms give it geometric content
Quaternionic fuzzy sets [Dai, 2023]	Min/max on quaternion-valued memberships	No t-norm, no residuum; compatible with but independent of this work
Lattice-ordered groups [Kopytov & Medvedev, 1994]	Lattice + group; non-abelian examples exist on free groups	$\mathbb{H}^\times$ cannot be lattice-ordered: Theorem 24 blocks the positive cone. Our monoid is not a group (B^4 has no inverses for $\lvert q\rvert < 1$), so the l-group theory does not directly apply

10.3 Scope of the obstruction

Theorem 31 eliminates six families of orders. The relationship between these eliminations and the sharp boundary of Theorem 30 is characterised in Remark 33: the trilemma holds precisely under the equal-norm condition $|a| = |b|$. Three further observations sharpen the scope.

Proposition 37 (No finite closed NC domain)**.** *Let $D \subset \mathrm{int}(B^4) \cup \{\mathbf{0}, \mathbf{1}\}$ be finite and closed under Hamilton product. Then $D \subseteq \{\mathbf{0}, \mathbf{1}\}$.*

Proof. Suppose $d \in D \setminus \{\mathbf{0}, \mathbf{1}\}$. Then $d \in \mathrm{int}(B^4)$, so $0 < |d| < 1$. The sequence $d, d^2, d^3, \ldots$ lies in D by closure, and $|d^n| = |d|^n$ (norm multiplicativity of $\mathbb{H}$). Since $0 < |d| < 1$, the norms $|d|^n$ are pairwise distinct, so the elements d^n are pairwise distinct—an infinite subset of the finite set D, a contradiction. □

Consequently, any finite domain containing non-commuting elements a, b with $|a|, |b| < 1$ is not closed under the Hamilton product, so the operation $T\colon D \times D \to D$ is not well-defined. Combined with the Q_8 impossibility (Theorem 25), which handles the boundary case $|q| = 1$: no finite domain supports the trilemma triple.

A much stronger result holds for the natural domain $V = [0, 1] \times [-1, 1]^3$ of the quaternionic logic.

Theorem 38 (Angular accumulation)**.** *Let $q \in V$ with $\mathrm{Im}(q) \neq 0$. Then there exists a finite N such that $q^N \notin V$.*

Proof. Write $q = |q|(\cos\theta + \sin\theta\, \hat{u})$ where $\hat{u} = \mathrm{Im}(q)/|\,\mathrm{Im}(q)|$ is a unit pure imaginary quaternion and $\theta = \arccos(\mathrm{Re}(q)/|q|) \in (0, \pi/2]$ (the bounds follow from $\mathrm{Re}(q) \geq 0$ and $|\,\mathrm{Im}(q)| > 0$). By De Moivre's theorem for quaternions ($\hat{u}^2 = -1$ gives the same exponential structure as complex numbers):

$$q^n = |q|^n(\cos(n\theta) + \sin(n\theta)\, \hat{u}).$$

In particular, $\mathrm{Re}(q^n) = |q|^n \cos(n\theta)$.

Since $\theta > 0$, there exists a smallest N with $N\theta > \pi/2$; explicitly, $N \leq \lceil \pi/(2\theta) \rceil + 1$. At $n = N$: if $\cos(N\theta) < 0$, then $\mathrm{Re}(q^N) < 0$, so $q^N \notin V$. If $\cos(N\theta) = 0$ (i.e. $N\theta = \pi/2$ exactly), then $\cos((N+1)\theta) = -\sin\theta < 0$, so $q^{N+1} \notin V$. □

Corollary 39 (Only commutative sub-monoids of V)**.** *Every sub-monoid of $(V, \cdot_{\mathbb{H}})$ contained in V is commutative. That is: if $D \subseteq V$ is closed under the Hamilton product and contains $\mathbf{1}$, then $ab = ba$ for all $a, b \in D$.*

Proof. Suppose $d \in D$ has $\mathrm{Im}(d) \neq 0$. By closure, $d^n \in D \subseteq V$ for all $n \geq 1$. But Theorem 38 gives $d^N \notin V$ for some finite N, a contradiction. So every $d \in D$ is real: $D \subseteq \{(r, 0, 0, 0) : r \in [0, 1]\}$. Real quaternions commute. □

Theorem 40 (Quaternionic trilemma for V-domains)**.** *Let $D \subseteq V$ contain $\mathbf{0}$ and $\mathbf{1}$, and let $T\colon D \times D \to D$ be the Hamilton product restricted to D. Then at most two of {(a) associativity, (b) non-commutativity, (c) compatible lattice order} hold.*

Proof. Condition (c) requires $T\colon D \times D \to D$ to be well-defined, hence D must be closed under T. Corollary 39 then forces D to be commutative, contradicting (b). □

The trilemma is therefore a *theorem* on the natural domain V of the quaternionic logic. The mechanism is angular entropy production: each non-real element carries an angle $\theta = \arccos(\mathrm{Re}(q)/|q|) > 0$, and iterated multiplication accumulates this angle linearly ($n\theta$). The boundary $\mathrm{Re} \geq 0$ of V can absorb at most $\pi/2$ radians of angular entropy; beyond this threshold the system exits V. The non-commutativity and the boundary violation share the same algebraic root: the relation $\hat{u}^2 = -1$ for pure imaginary unit quaternions.

A stronger result holds for the full ball B^4, where powers remain in B^4 and the angular accumulation argument does not apply. The obstruction shifts from closure failure to a lattice-theoretic incompatibility within the non-commutative sub-monoid itself.

Proposition 41 (Level-2 incomparability in B^4)**.** *Let $r \in (0,1)$, $a = ri$, $b = rj$, and let $\leq$ be any partial order on $M = \langle a, b\rangle \cup \{\mathbf{0}, \mathbf{1}\}$ with $\mathbf{0}$ as bottom, $\mathbf{1}$ as top, and the Hamilton product monotone on both sides. Then:*

(a) *$a \| b$ (the generators are incomparable);*

(b) *a^2, ab, and ba are pairwise incomparable.*

Proof. **(a)** Suppose $a \leq b$. Left-multiply by a: $a^2 \leq ab$, i.e. $-r^2 \leq r^2k$. Right-multiply by b: $ab \leq b^2$, i.e. $r^2k \leq -r^2$. Together: $r^2k = -r^2$, which is false. The case $b \leq a$ is symmetric (yielding $-r^2k = -r^2$).

(b) We show the case $a^2 \leq ab$ in detail; the remaining five are analogous.

Suppose $a^2 \leq ab$. Left-multiply by a: $a^3 \leq a^2b$, i.e. $-r^3i \leq -r^3j$. Right-multiply by b: $a^2b \leq ab^2 = a^3$, i.e. $-r^3j \leq -r^3i$. By antisymmetry, $-r^3i = -r^3j$, contradicting $i \neq j$. In each of the six cases, two applications of monotonicity (one left, one right multiplication by generators) produce a cycle $p \leq q \leq p$ between distinct level-3 quaternions. The full case analysis is verified computationally in the supplementary script `b4_proof_verify.py`. □

Theorem 42 (Lattice obstruction for B^4)**.** *Let $\hat{u}, \hat{v}$ be orthogonal unit purely imaginary quaternions and $r \in (0,1)$. The sub-monoid $M = \langle r\hat{u}, r\hat{v}\rangle \cup \{\mathbf{0}, \mathbf{1}\}$ of $(B^4, \cdot_{\mathbb{H}})$ admits no partial order $\leq$ satisfying simultaneously:*

(i) *$\mathbf{0}$ is bottom and $\mathbf{1}$ is top;*

(ii) *$x \leq y$ implies $zx \leq zy$ and $xz \leq yz$ for all $z \in M$;*

(iii) *$(M, \leq)$ is a lattice.*

Proof. By SU(2) conjugation (which preserves norms, products, and the order structure), we may assume $\hat{u} = i$, $\hat{v} = j$, so $a = ri$ and $b = rj$. The products $a^2 = b^2 = -r^2$, $ab = r^2k$, $ba = -r^2k$ are pairwise distinct.

Forced bounds. From $0 \leq a, b \leq 1$ and (ii): for any word $w = w_1w_2 \cdots w_n$ with $w_i \in \{a, b\}$ and $n \geq 2$, induction on n using $w_n \leq 1$ gives $w \leq w_1w_2$. In particular, every element of M at level ≥ 3 is below its two-generator prefix in $\{a^2, ab, ba\}$.

Upper bounds of $\{a^2, ba\}$. We claim that the only elements of M above both a^2 and ba are $\{a, b, 1\}$.

Level 1: $a \geq a^2$ (from $a \leq 1$, multiply by a) and $a \geq ba$ (from $b \leq 1$, right-multiply by a). Similarly $b \geq a^2$ (since $a^2 = b^2 \leq b$) and $b \geq ba$ (from $a \leq 1$, left-multiply by b).

Level 2: any of $ab \geq a^2$, $ba \geq a^2$, or $a^2 \geq ba$ contradicts Proposition 41(b).

Level ≥ 3: if $w \geq a^2$ for a word of length $n \geq 3$, then $a^2 \leq w \leq w_1w_2$, forcing $a^2 \leq w_1w_2$. Proposition 41(b) excludes $w_1w_2 \in \{ab, ba\}$, so $w_1w_2 = a^2$, giving $w = a^2$ by antisymmetry. But $|w| = r^n < r^2 = |a^2|$ for $n \geq 3$, so $w \neq a^2$—a contradiction.

$\mathbf{0}$: if $0 \geq a^2$ then $a^2 = 0$, contradicting $a^2 = -r^2 \neq 0$.

No join. The upper bounds of $\{a^2, ba\}$ in M are exactly $\{a, b, 1\}$. Both a and b are minimal among these ($a < 1$, $b < 1$, but $a \| b$ by Proposition 41(a)). Two incomparable minimal upper bounds admit no least upper bound. The join $a^2 \vee ba$ does not exist, so $(M, \leq)$ is not a lattice. □

The obstruction mechanism differs fundamentally from the V-domain case. In V, non-commutativity is eliminated entirely (every sub-monoid is commutative). In B^4, non-commutative sub-monoids exist—the Hamilton product is closed on B^4—but they cannot carry a compatible lattice order. The root cause is the same in both cases: the relation $\hat{u}^2 = -1$ for imaginary unit quaternions.

The equal-norm hypothesis is sharp. When $|a| \neq |b|$, the collapse $a^2 = b^2$ that drives the cyclic swap at level 3 no longer holds, and compatible monotone lattice orders exist (Remark 33).

This delineates the precise boundary of the trilemma within B^4: non-commutativity and compatible order coexist exactly when the generators have unequal norms, restoring sufficient algebraic room for the lattice.

Remark 43 (Quaternionic specificity)**.** *The obstruction is specific to the quaternionic setting. The only normed division algebras over $\mathbb{R}$ are $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, and $\mathbb{O}$ (Hurwitz's theorem). Among these: $\mathbb{R}$ and $\mathbb{C}$ are commutative (no property (b)), $\mathbb{O}$ is non-associative (no property (a)), and $\mathbb{H}$ is the subject of this paper. Clifford algebras $\mathrm{Cl}(p,q)$ with $p+q \geq 3$ are not division algebras and thus admit zero divisors, precluding well-defined residuals. The trilemma is therefore a statement about the* unique *algebraic setting—$\mathbb{H}$—where all three properties can be meaningfully posed.*

Remark 44 (Residuation under the norm order)**.** *One might ask whether the division-based residuals (Definition 11), truncated to B^4 via $a \to_L b = \pi_{B^4}(a^{-1}b)$, preserve the adjunction approximately. Proposition 27 settles this: the only monotone order on B^4 compatible with the Hamilton product is the norm order $|q_1| \leq |q_2|$, which collapses the lattice to $[0,1]$ with the product t-norm. Under this order, the residual is $|a| \to |b| = \min(1, |b|/|a|)$ and the adjunction holds* exactly*—but the quaternionic structure beyond the norm is invisible. No intermediate approximation between full quaternionic residuation and norm collapse is available, because the monotonicity failures (Theorem 5) are structural, not perturbative.*

11 Conclusion

The Hamilton product of quaternions is a natural candidate for a non-commutative conjunction in quaternionic logic: it couples the four epistemic axes, satisfies bilateral identity and associativity, admits left and right implications with modus ponens, and reduces to standard commutative t-norms under every proper restriction. Its non-commutativity is emergent—a genuinely four-dimensional phenomenon absent from all lower-dimensional projections.

The obstructions to a full residuated lattice are equally deep. Six independent impossibility results eliminate every natural family of lattice orders on B^4: bi-invariant cones (SU(2) transitivity), finite group orders (Q_8 exhaustive), real-dominant subcones (analytical bound), norm-based orders (1D collapse), component-wise orders (closure failure), and inversion-closed domains (sphere exclusion). These impossibilities hold in the equal-norm regime. For unequal-norm generators, the obstruction dissolves: every pair of properties is individually achievable (Remark 32), and the triple is impossible precisely when $|a| = |b|$.

This elimination is not a deficiency of any particular construction—it traces to the defining relations $i^2 = j^2 = k^2 = -1$. The minus signs that produce non-commutativity are the same signs that break monotonicity—and, via De Moivre's theorem, the same signs that force iterated products out of the truth-value domain V.

The quaternionic trilemma is a theorem in two settings. For V-domains (Theorem 40), the proof is that every sub-monoid of V under the Hamilton product is commutative (Corollary 39), because the angular entropy $\theta = \arccos(\mathrm{Re}(q)/|q|)$ of any non-real element accumulates linearly under iteration, exceeding V's capacity $\pi/2$ in finitely many steps (Theorem 38). For equal-norm sub-monoids of B^4 (Theorem 42), the proof is that the collapse $a^2 = b^2$ forces pairwise incomparability of level-2 products and prevents the existence of joins. The equal-norm condition is sharp: unequal-norm sub-monoids admit compatible lattice orders (Remark 33), so the trilemma boundary is precisely the locus $|a| = |b|$. Six family eliminations (Theorem 31) and nine computationally surveyed candidates (Table 1) independently corroborate the impossibility in the equal-norm regime.

References

Ciungu, L.C. (2006). Classes of residuated lattices. *Annals of the University of Craiova, Mathematics and Computer Science Series*, 33, 56–70.

Dai, S. (2023). Quaternionic fuzzy sets. *Axioms*, 12(5), 490.

Esteva, F. & Godo, L. (2001). Monoidal t-norm based logic: towards a logic for left-continuous t-norms. *Fuzzy Sets and Systems*, 124(3), 271–288.

Galatos, N., Jipsen, P., Kowalski, T. & Ono, H. (2007). *Residuated Lattices: An Algebraic Glimpse at Substructural Logics*. Elsevier.

Hájek, P. (1998). *Metamathematics of Fuzzy Logic*. Kluwer.

Hájek, P. (2003). Observations on non-commutative fuzzy logic. *Soft Computing*, 7(1), 38–43.

Kopytov, V.M. & Medvedev, N.Ya. (1994). *The Theory of Lattice-Ordered Groups*. Kluwer.

Lambek, J. (1958). The mathematics of sentence structure. *American Mathematical Monthly*, 65(3), 154–170.

Mulvey, C.J. (1986). &. *Rendiconti del Circolo Matematico di Palermo, Serie II, Supplemento*, 12, 99–104.

Ornelas Brand, J.A. (2026). Quaternionic Logic: A G-Lattice Unifying Boolean and Fuzzy Frameworks. Manuscript in preparation.

Restall, G. (2000). *An Introduction to Substructural Logics*. Routledge.